



ADAMA SCIENCE AND TECHNOLOGY UNIVERSITY

SCHOOL OF ELECTRICAL ENGINEERING AND COMPUTING

DEPARTMENT OF ELECTRICAL POWER AND CONTROL ENGINEERING

POWER SYSTEMS 1 (EPCE 3206)

INSTRUCTOR :- Mekdes Gemechu

CHAPTER – ONE

FUNDAMENTALS OF POWER SYSTEM



Outline

✓ Introduction

✓ Generation

✓ Transmission

✓ Distribution

✓ AC and DC Transmission

✓ Single Phase and Three Phase Transmission

INTRODUCTION

Advantage of Electric Energy over other form of energy

Electric Energy is:-

- **Flexible**: Easily converted to light, heat, mechanical energy...
- **Cost**: it is cheaper
- **Transportation**: easy & simple to transport, cable
- **Clean**: has no smoke

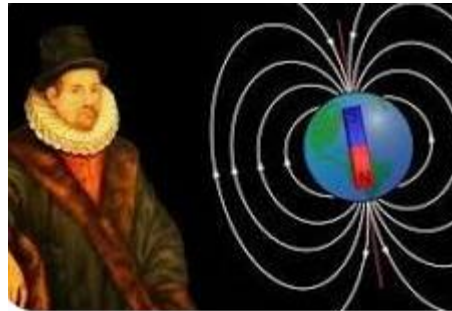
What is power system?

Power system: is a science which deals with:

- Generation
- Transmission
- Distribution and
- Utilization of electric power system.

Historical development of power system

- **William Gilbert (1544-1603)** Gilbert took the decisive step in the history of science from occasional experimental investigations of isolated problems by his predecessors to a **comprehensive examination of a whole area of physics, magnetism, and the creation of a qualitative theory employing genuine theoretical concepts**



- Sir **Joseph John Thomson** OM FRS (18 December 1856 – 30 August 1940) was a British physicist and Nobel Laureate in Physics. J.J. Thomson's experiments with cathode ray tubes showed that **all atoms contain tiny negatively charged subatomic particles or electrons**. Thomson proposed the plum pudding model of the atom, which had negatively-charged electrons embedded within a positively-charged "soup."

....History

- **Alessandro Volta invented the voltaic pile**, the first electric battery.
- With this invention Volta proved that **electricity could be generated chemically** and debunked the prevalent theory that electricity was generated solely by living beings. Volta's invention sparked a great amount of scientific excitement and led others to conduct similar experiments, which eventually led to the development of the field of electrochemistry
- **Michael Faraday**
- in 1821 he invented **the first electric motor**,
- In the early 1830s he discovered a way to convert mechanical energy into electricity on a large scale, creating the **first electric generator**

....History

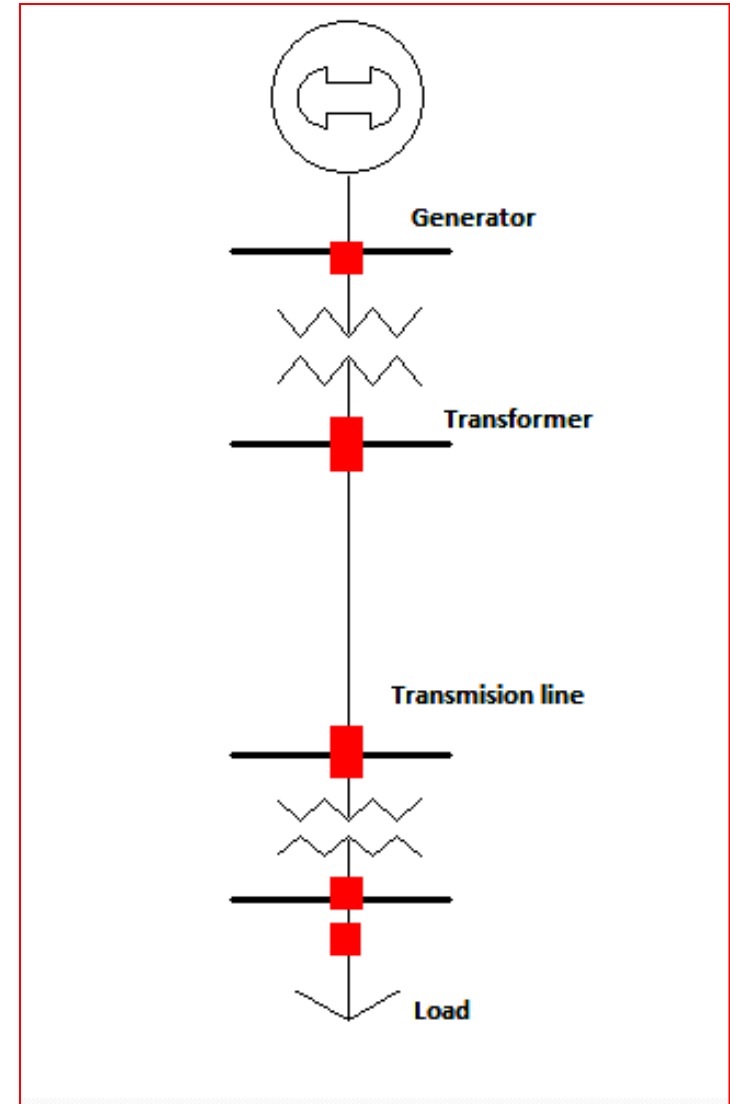
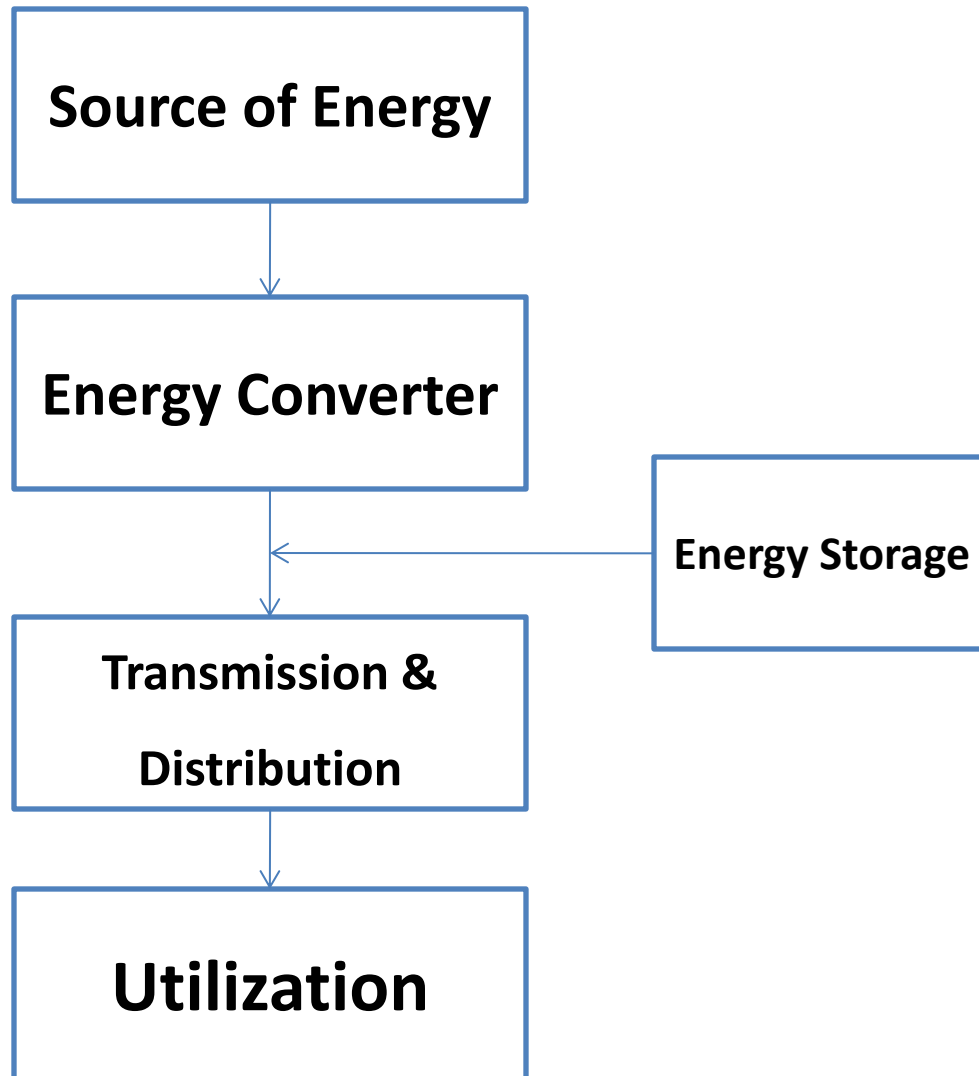
Thomas Alva Edison

- One of the most famous and prolific inventors of all time, Thomas Alva Edison exerted a tremendous influence on modern life, contributing inventions such as the **incandescent light bulb**.
- **Nikola Tesla, (born July** 9/10, 1856, Smiljan, Austrian Empire [now in Croatia]—died January 7, 1943, New York, New York, U.S.), Serbian American inventor and engineer who **discovered and patented the rotating magnetic field**, the basis of most **alternating-current machinery**.

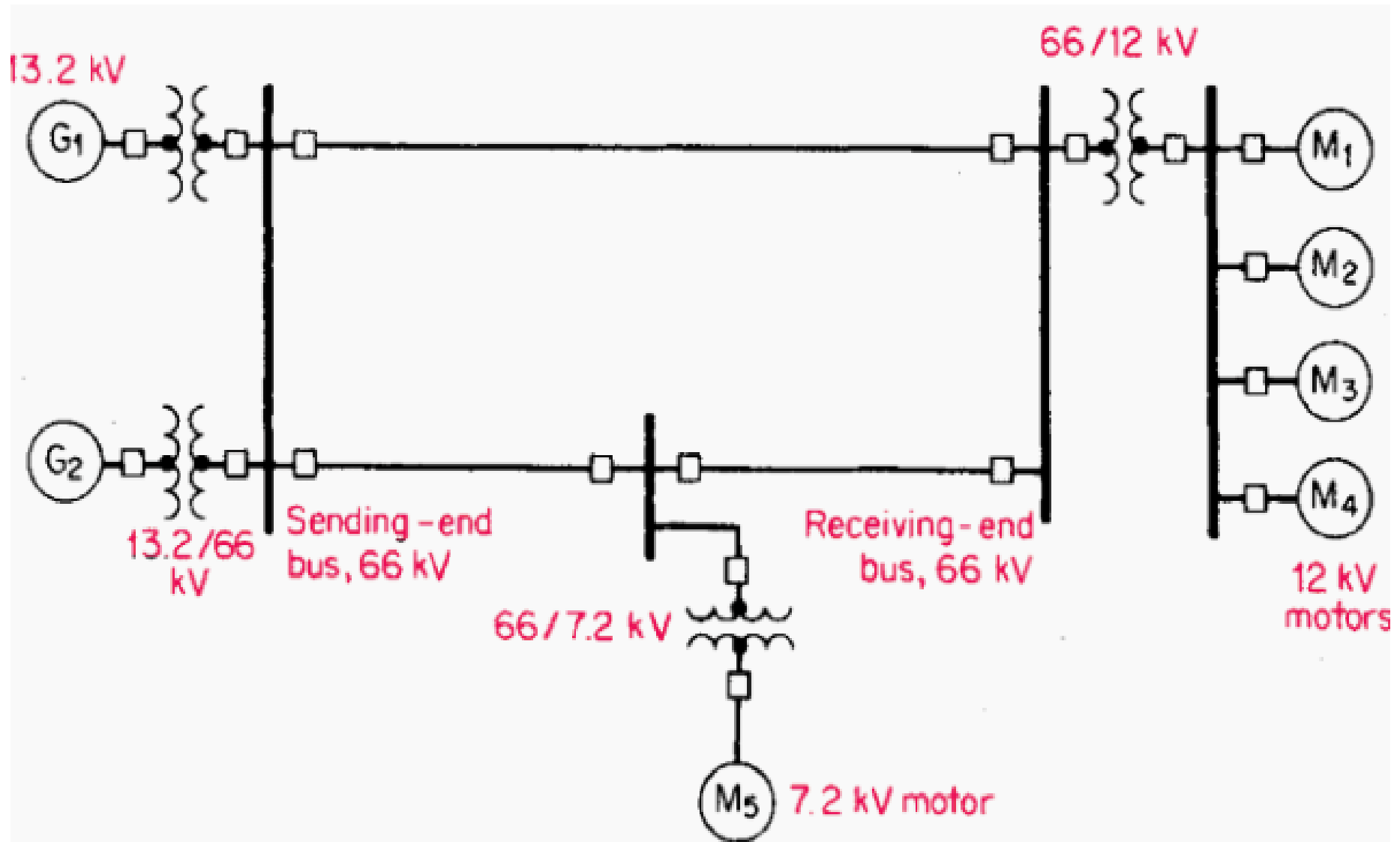
....History

- **James Watt (1736-1819)** Famous for: Inventing **the Watt steam engine**, which converted steam back to water. **Developing a rotary engine** which mechanised weaving, spinning and transport.
- On **September 30, 1882**, the world's first hydroelectric power plant began operation on the Fox River in Appleton, Wisconsin. The plant, later named the Appleton Edison Light Company, was initiated by Appleton paper manufacturer H.J.
- Transformer:- Takara Hasbro in 1885 invented transformer.

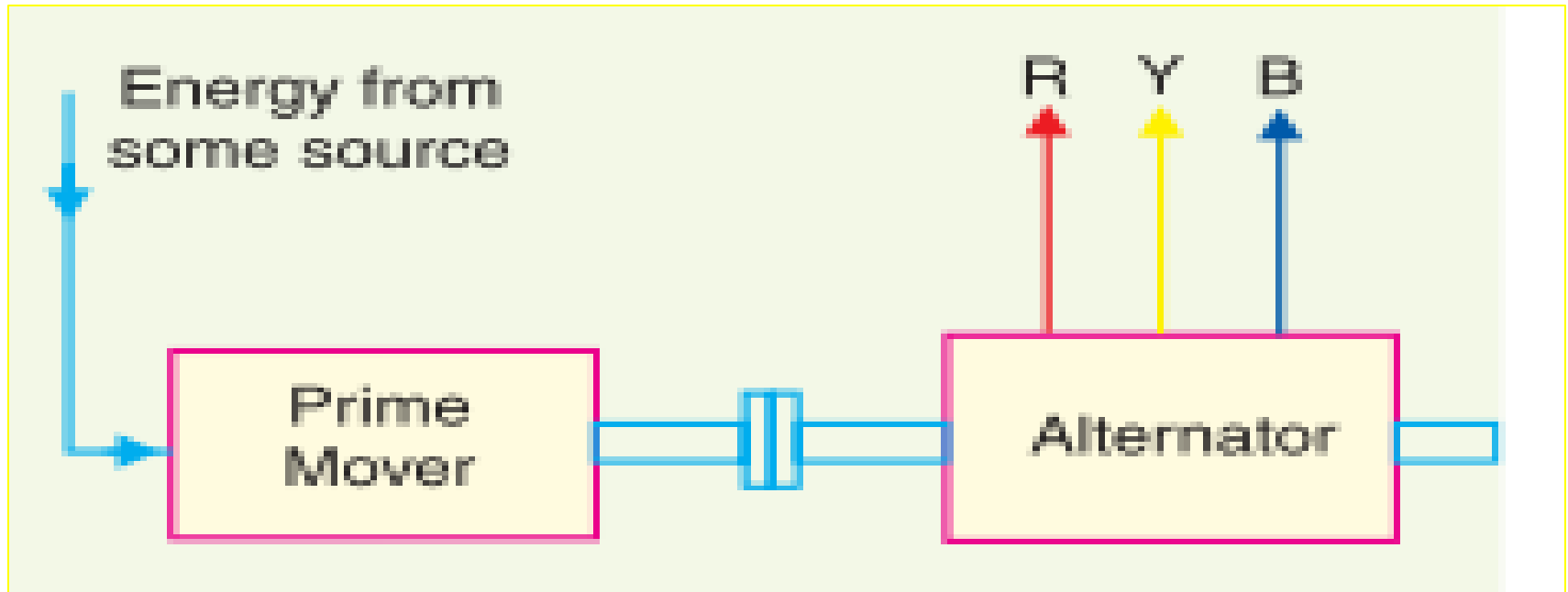
Structure of power system



Structure of power system



1.Generation



- ✓ Generators can be classified as AC/ DC generators
- ✓ **Synchronous generator** are the most common used AC Generators.
- ✓ The max. capacity of a single generator ranges up to 34KV, due to insulation problems.

Energy Sources

1. Conventional Energy sources

✓ **Non Renewable Energy Sources**

✓ Convectional energy sources are *stocks of energy sources* which **deplete** (*cannot continuously re emerged*) using natural process.

✓ Some of convectional Energy sources are:-

1. *Fossil fuels:- from remains of dead animals and plants*

✓ Coal - Solid Fuel

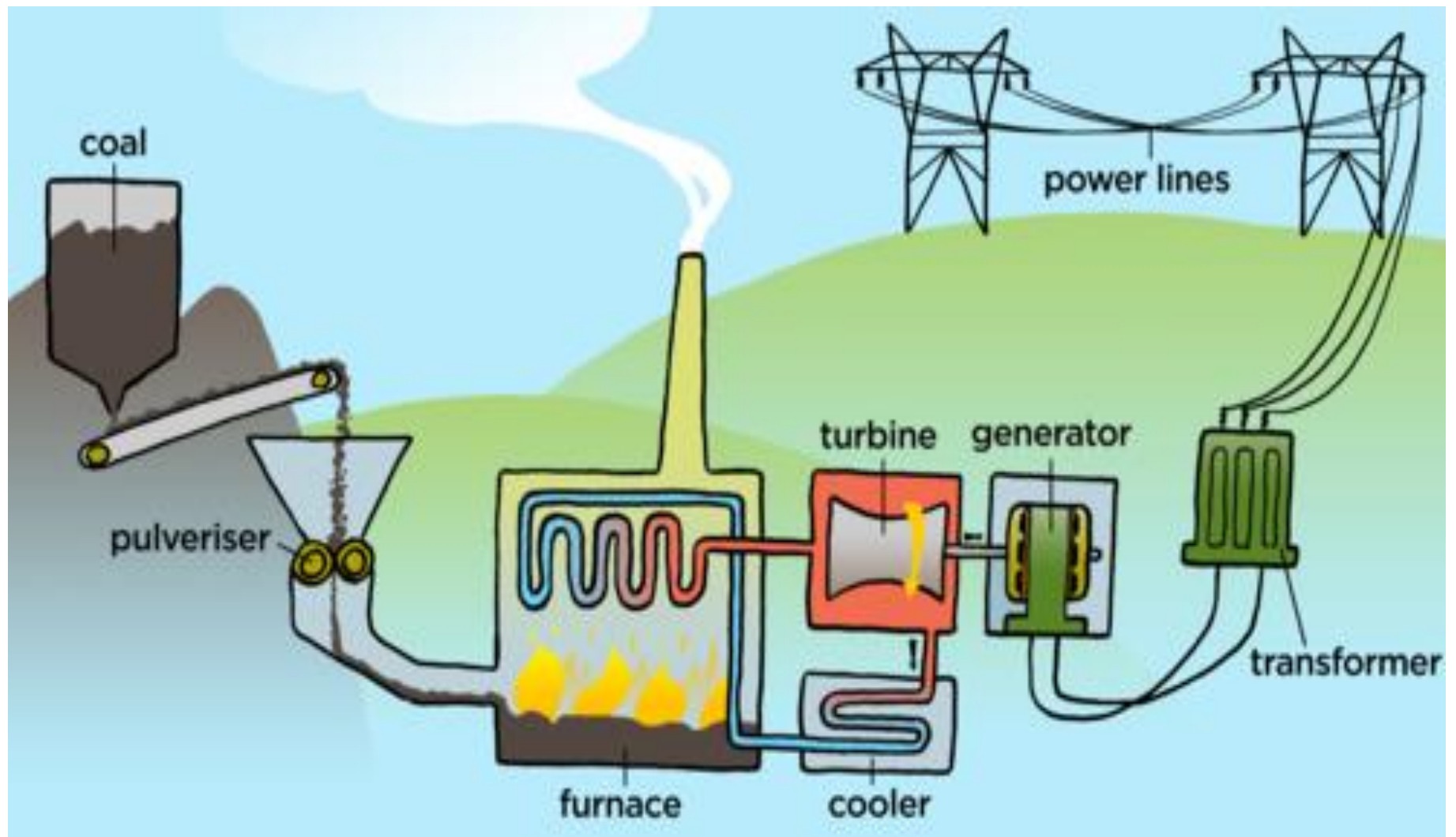
✓ Oil – Liquid fuel &

✓ Gas – as Natural Gas

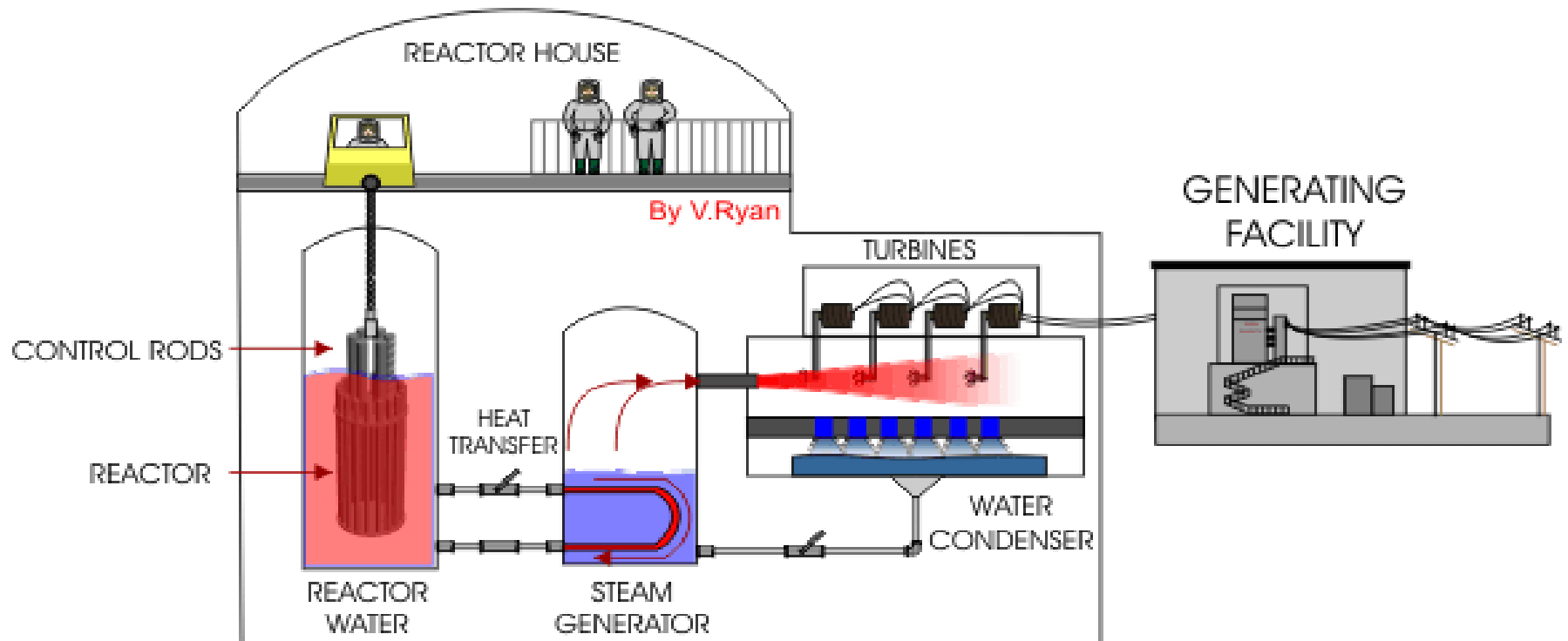
2. *Nuclear fuels:- by fission of uranium*

Refer on text books

Coal



Nuclear Power Plant



In **nuclear power plants**, neutrons collide with uranium atoms, splitting **them**. ...

In the core of **nuclear** reactors, the fission of uranium atoms **releases energy** that heats water to about **520 degrees Fahrenheit**.

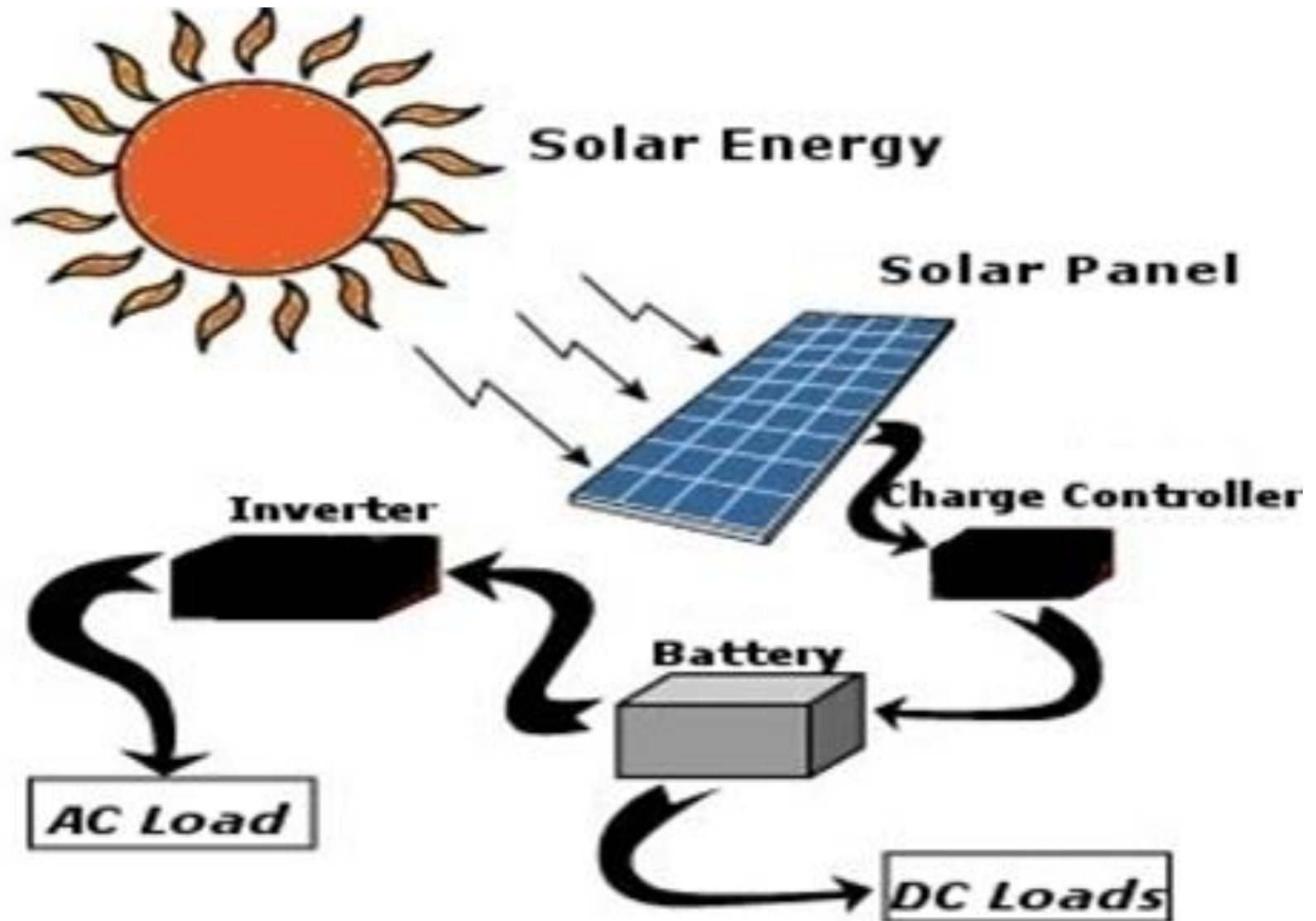
This **hot water** is then used to **spin turbines** that are connected to generators, producing electricity.



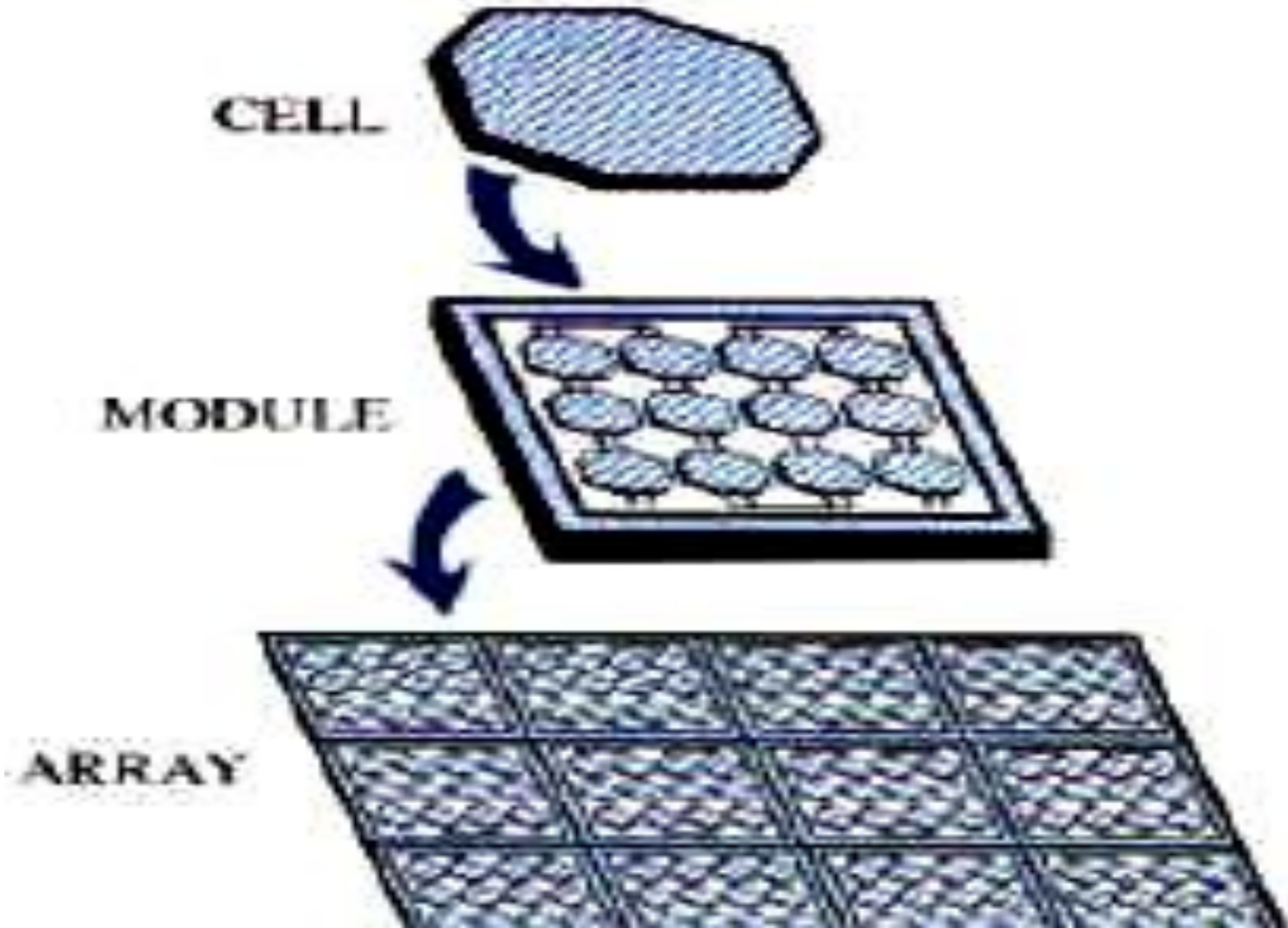
B. Non-Conventional Energy Sources

- Non-conventional energy sources also called **Renewable energy**.
- They are sources that are continuously **replenished** by natural processes.
- *Some sources of renewable energy are:-*
 - sunlight (solar Energy)
 - wind
 - falling-water
 - sea-waves (tidal)
 - geothermal heat
 - biomass

Solar Photovoltaic (PV):



Solar Photovoltaic (PV):





Wind Energy Technology

1. **Vertical - axis wind** turbines, in which the axis of rotation is vertical with respect to the ground (and roughly perpendicular to the wind stream),
2. **Horizontal-axis turbines**, in which the axis of rotation is horizontal with respect to the ground (and roughly parallel to the wind stream.)

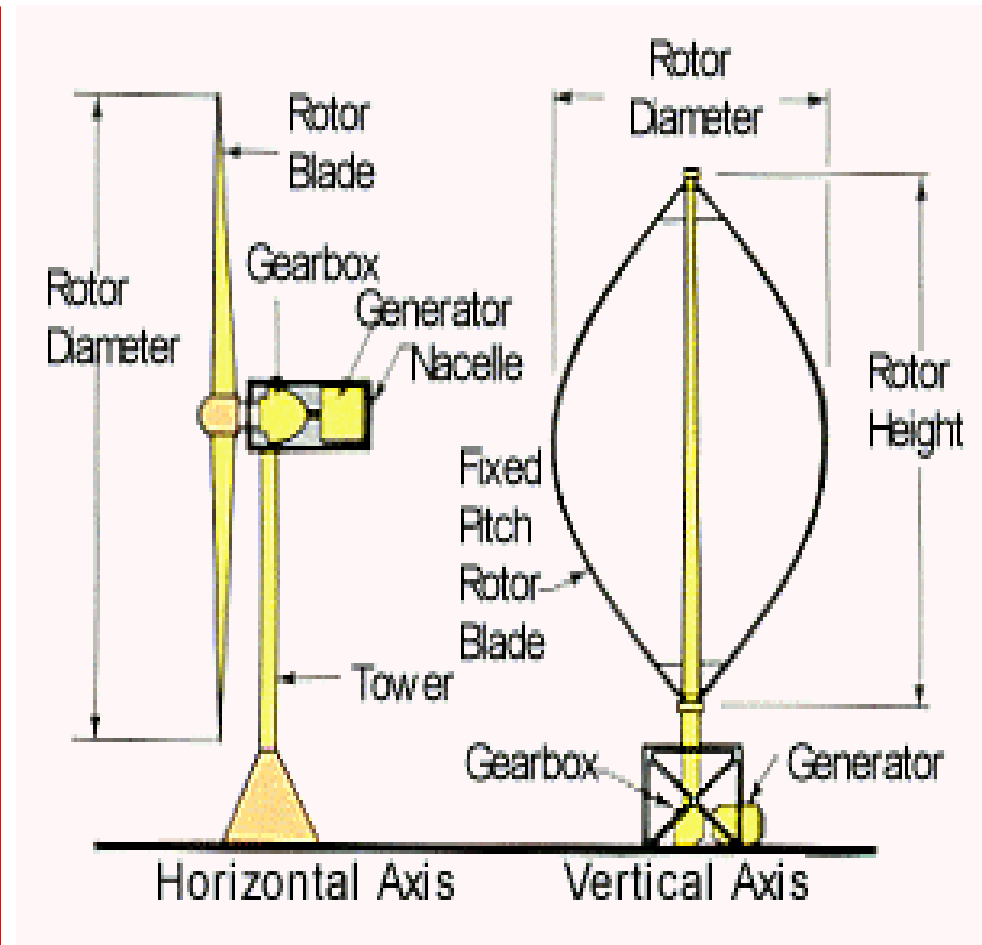
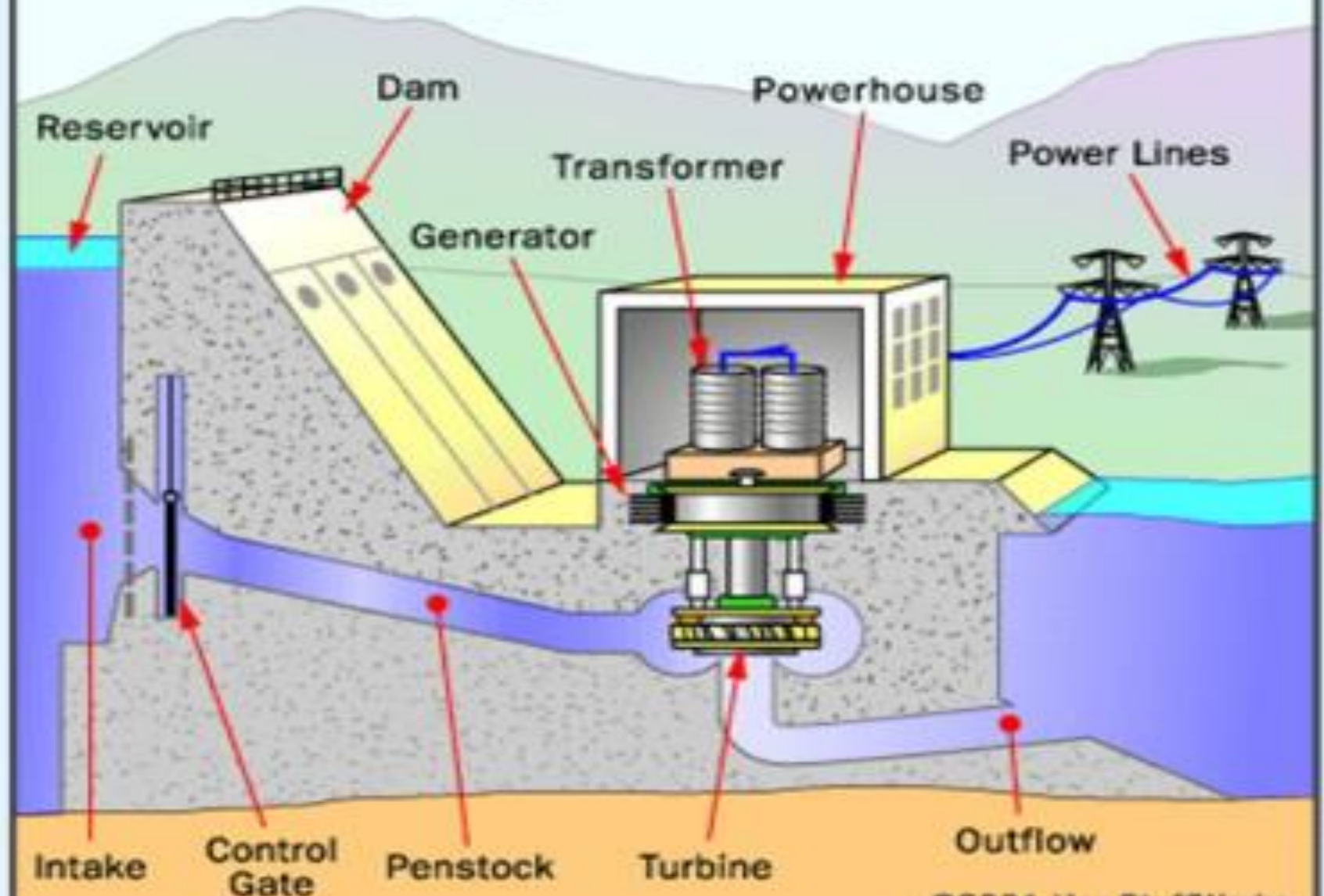
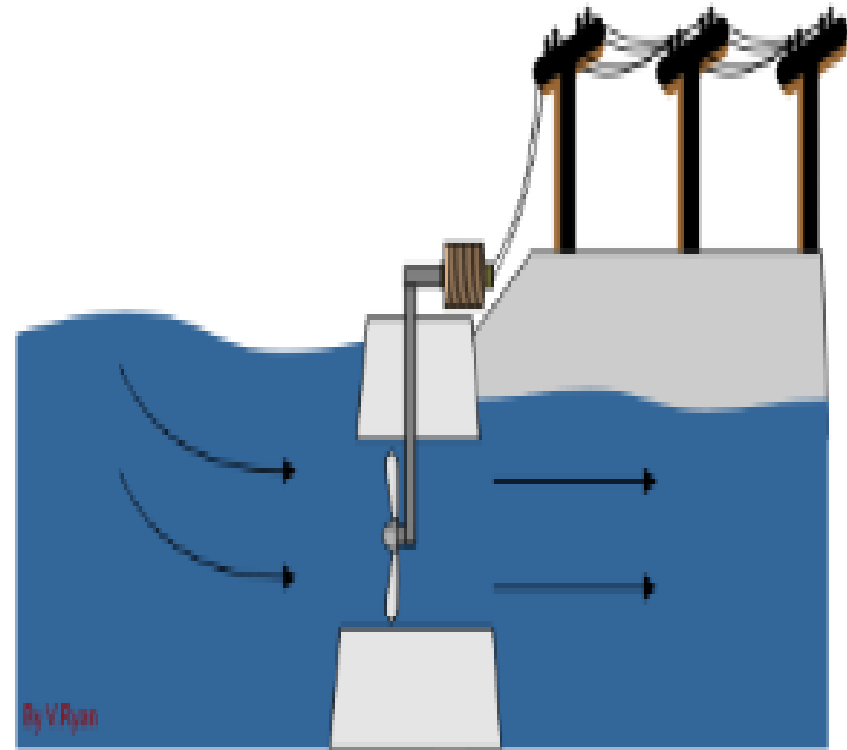


Figure 1.5 Wind Turbine Configuration

Inside a Hydropower Plant



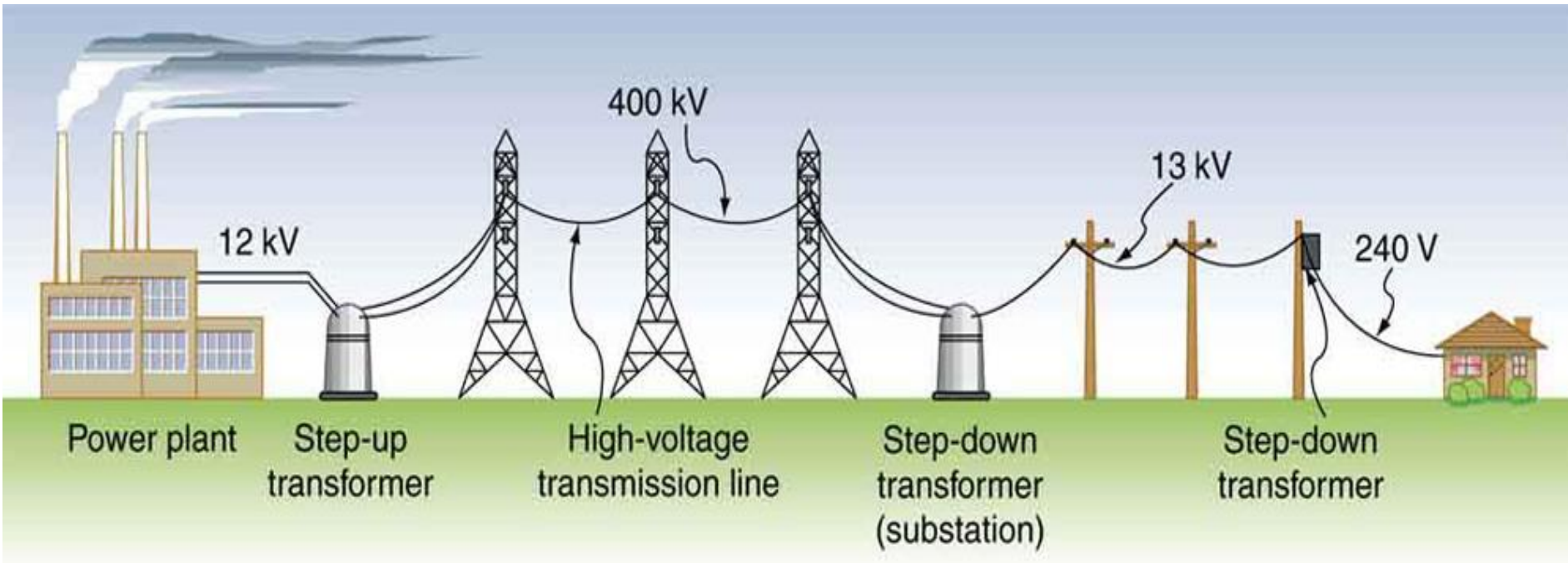
Tidal Energy



Tidal energy is created using the movement of **tides** and oceans, where the intensity of the water from the rise **and** fall of **tides** is a form of kinetic **energy**. **Tidal power** surrounds gravitational hydropower, which uses the movement of water to push a turbine to **generate electricity**

Refer on text books

Transformer



Transformer

- ✓ is a static machine
- ✓ operates on ac supply and works on the principle of mutual induction
- ✓ Transformers can be categorized in **different ways**, depending upon **their purpose**.



- The usual generation voltages are: 33KV, 11KV, 6.6kv...
- The **large network between power station and consumers** can be classified into four parts.

These are:-

- i. Primary transmission line : 760kv, 400kv, 230kv, 132KV
- ii. Secondary transmission line: 66kv, 33kv
- iii. Primary distribution: 11kv, 6.6kv
- iv. Secondary distribution: 380V, 220V

2. Transmission Systems

❑ The transmission system is composed of

✓ Step-up and Step-down substations

❑ Transmission lines can be

✓ DC or AC transmission system

➤ DC Transmission

❑ **Advantages.**

✓ It requires only two conductors

✓ There is no inductance, capacitance

✓ There is no reactance, phase displacement and surge problems

✓ less voltage drop so it has better voltage regulation.

DC Transmission

- ✓ The potential stress on the insulation is less it requires less insulation.
- ✓ Less corona loss and reduced interference with communication circuits.
- ✓ There are no stability problems and synchronizing difficulties.

DC Transmission

❑ Disadvantages

- ✓ Electric power cannot be generated at high d.c. voltage due to **commutation** problems.
- ✓ The d.c. voltage cannot be stepped up for transmission of power at high voltages.
- ✓ The d.c. switches and circuit breakers have their own limitations.

HVDC TRANSMISSION

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AC Transmission

- ❖ Now a day, electrical energy is almost exclusively generated transmitted and distributed in AC forms.

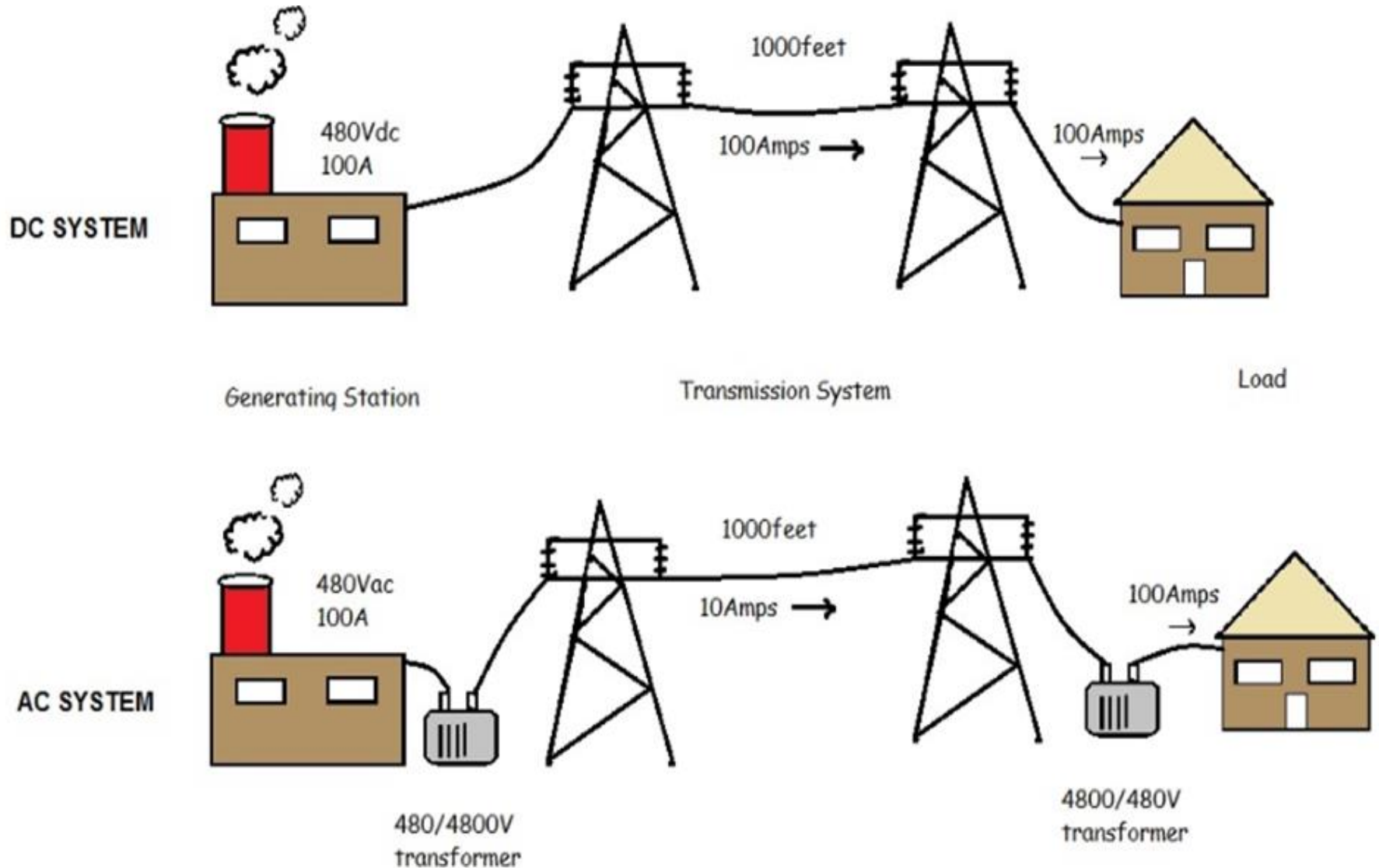
Advantages

- ✓ The power can be generated at high voltages.
- ✓ The maintenance of AC sub-stations is easy and cheaper.
- ✓ stepped up or stepped down by transformers with ease and efficiency.

Disadvantages

- ✓ Required more copper
- ✓ more complicated construction
- ✓ AC. line has capacitance. Therefore, there is a continuous loss of power due to charging current even when the line is open.

AC vs DC



Single phase and three-phase transmission

➤ Single phase system

- ✓ There will be only one phase,
- ✓ Minimum amount of power can be transported.
- ✓ Generating station and load station single phase

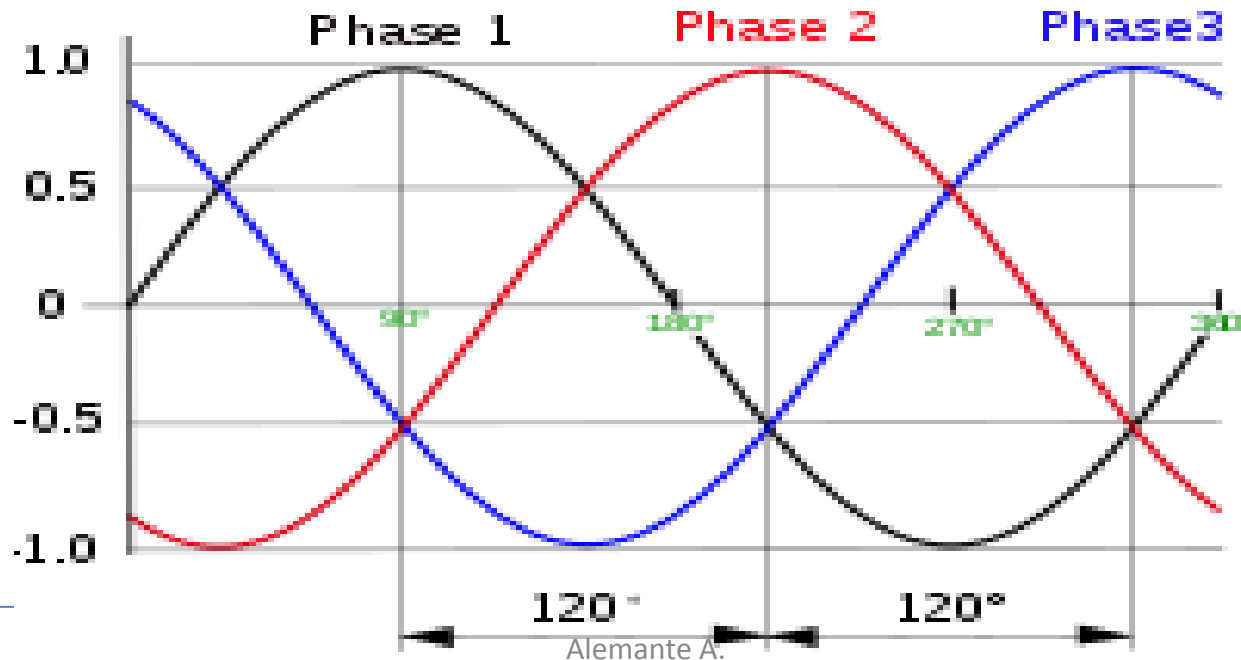


Pulsating power due to single phase supply

Three-phase transmission

➤ Three phase system

- ✓ Three phase system is where three phases are send together from the generator to the load.
- ✓ Each phase are having a phase difference of 120° , i.e. 120° angle electrically.
- ✓ Each of the three phases can be used as single phase.



Phase Shift & Operator

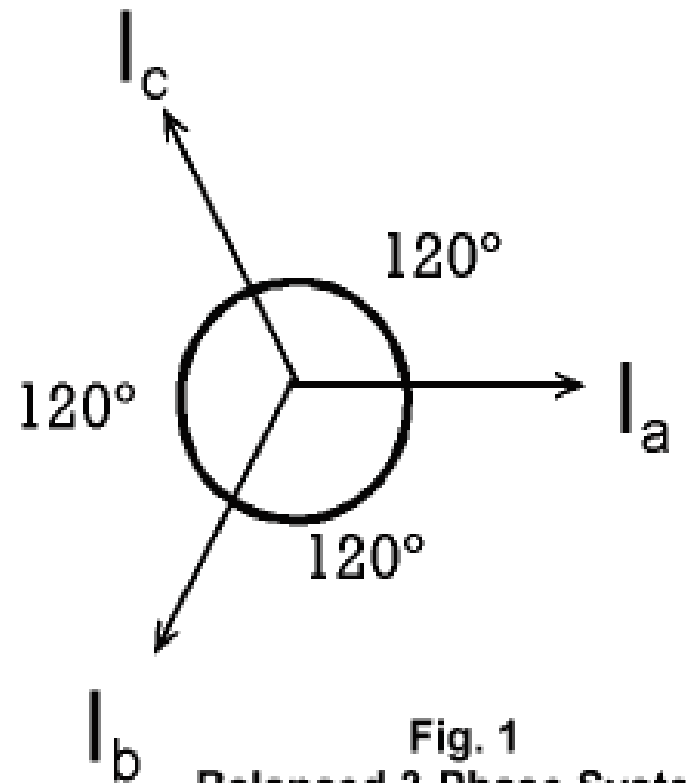
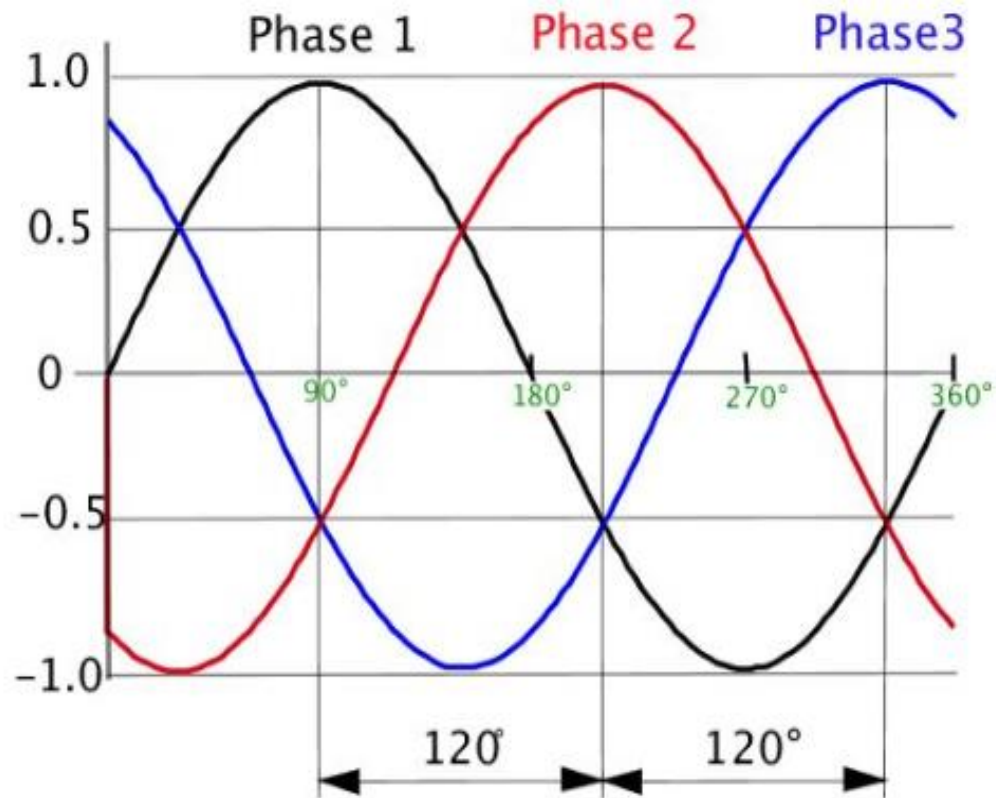
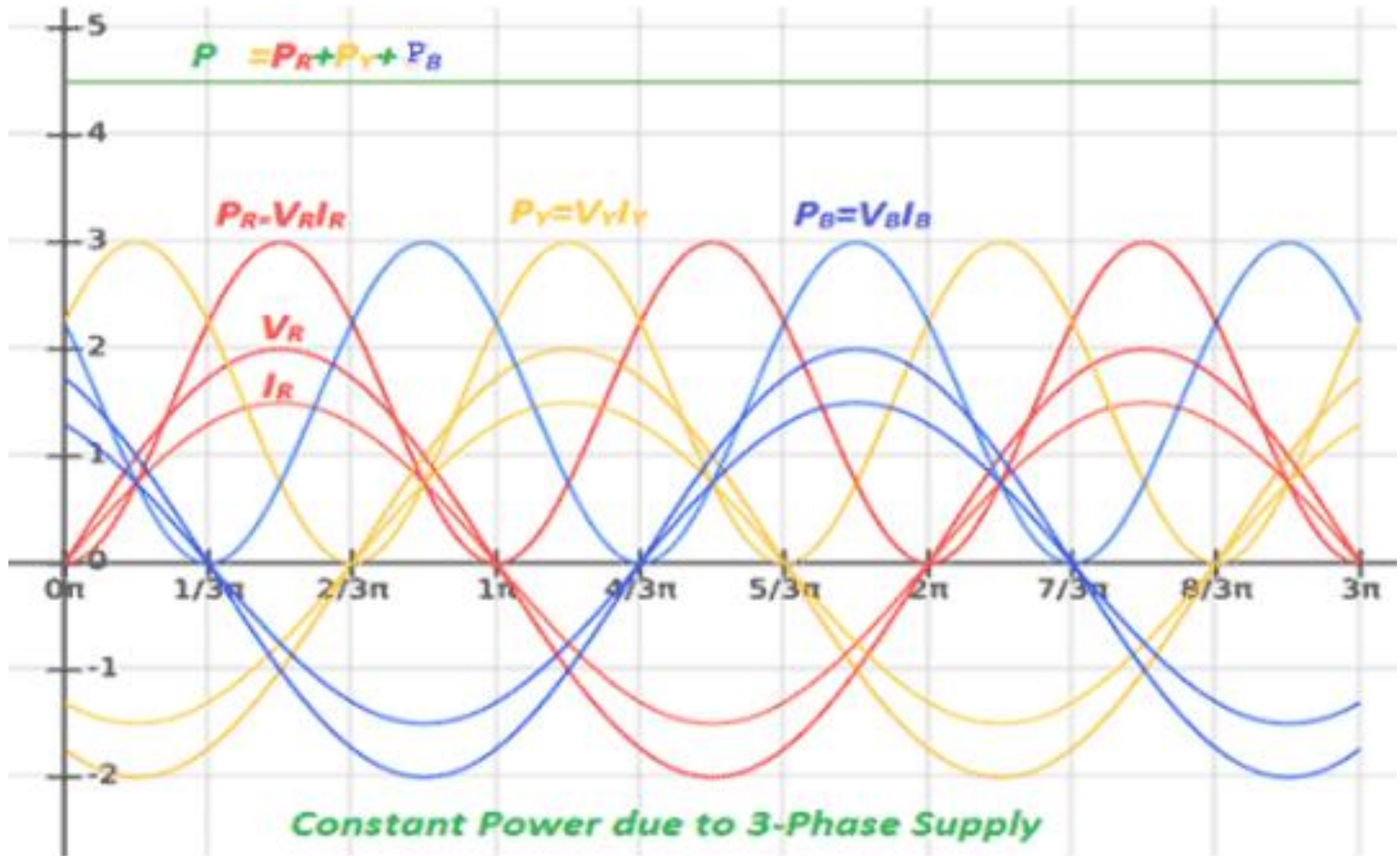


Fig. 1
Balanced 3-Phase System

Constant power due to 3-phase system



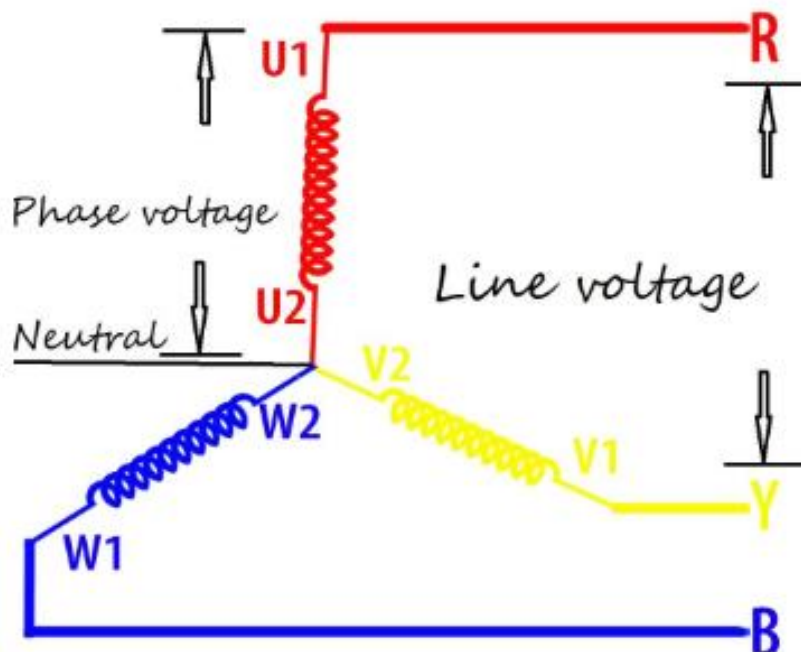
Why Three Phase is preferred Over Single Phase?

- ✓ Higher **power/weight** ratio of alternators smaller, lighter and cheaper.
- ✓ 3-phase requires **less copper** or aluminum to **transmit the same quantity of power** of a specific distance than a single phase system.
- ✓ Three **phase motors are self-starting** due to the rotating magnetic .
- ✓ But a single phase motor is not self-starting, it requires a capacitor and an auxiliary winding.
- ✓ In Single phase systems, **the instantaneous power is not constant** and is **sinusoidal** where as in a three phase the instantaneous power is always the same.
- ✓ Three phase supply can be **rectified into dc** supply with a lesser ripple factor.

Three phase circuit connections

Star Connection:-

- ✓ Star connection is preferred for **long distance power transmission** because it is having the neutral point.
- ✓ **Balanced system:-** equal current will flow through all the three phases - $I_R + I_Y + I_B = 0$
- ✓ When the current will not be equal in any of the phase, then it is **unbalanced current** -



$$V_{ph} = V_L / \sqrt{3} \quad \text{and} \quad I_{ph} = I_L$$

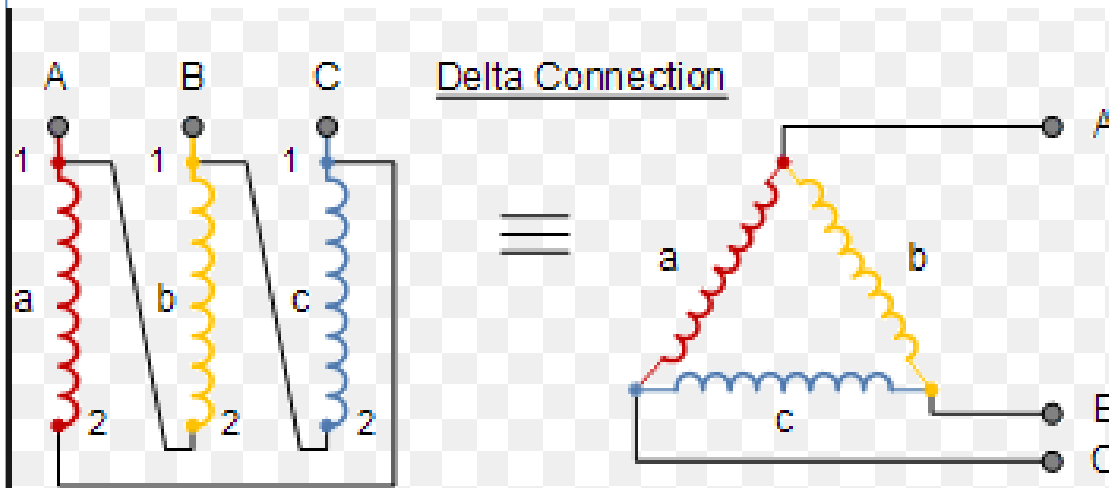
$$P = 3 \times V_{ph} I_{ph} \cos \phi$$

$$P = \sqrt{3} V_L I_L \cos \phi$$

total active power = 3 × phase power

Delta Connection

- ✓ Three wires alone and **no neutral terminal** is taken.
- ✓ Preferred for **short distance transmission** due to the problem of **unbalanced current** in the circuit.



$$V_{ab} = V_{an} \text{ or } V_L = V_{ph}$$

$$I_{ab} = \sqrt{3}I_{an} \text{ or } I_L = \sqrt{3}I_{ph}$$

$$P = 3 \times V_{ph} I_{ph} \cos \phi$$

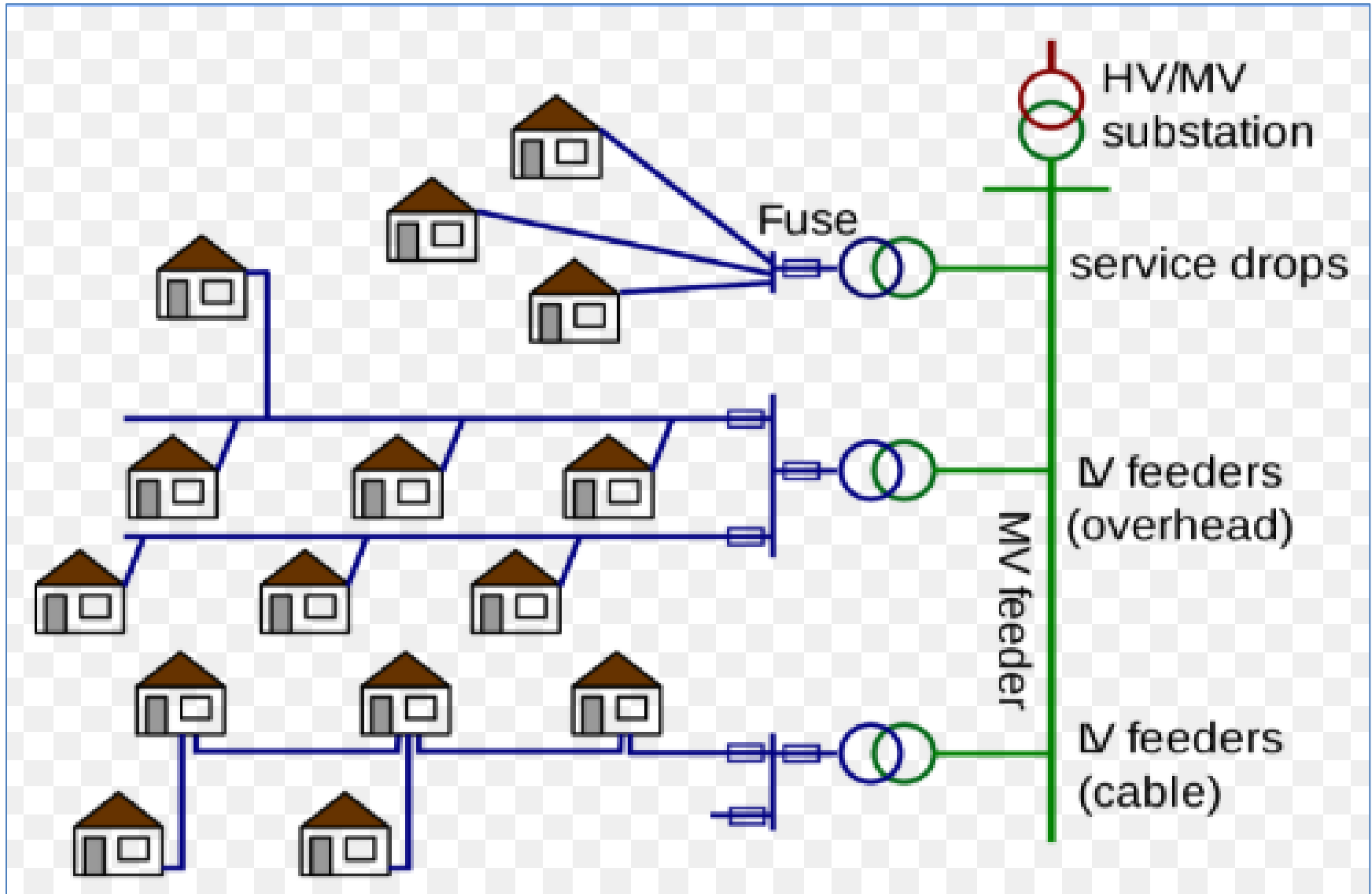
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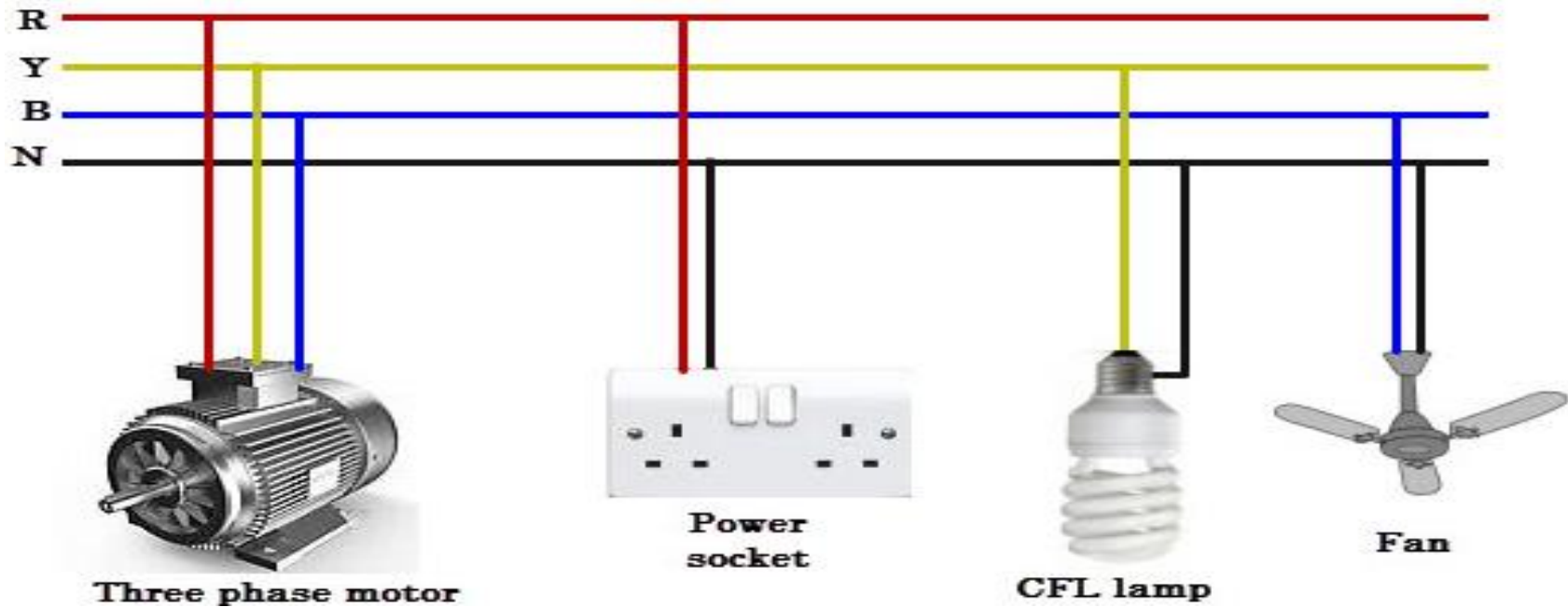
Distribution Systems

- Electric power distribution is the final stage in the delivery of electric power : it carries electricity from the transmission system to individual consumers.
- It is composed of:
 - ✓ Medium Voltage (MV) lines (33kV, 15 kV).
 - ✓ MV/LV transformer substations (33/15/ kV to 400/380 V).
 - ✓ 380,V 3-phase and 220 v single phase Distribution lines .

Distribution Systems



Load or Energy Sink



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Load or Energy Sink

- ✓ Load is the **end equipment** of the power system where the transmitted electrical energy is **converted to other forms** of useful energy.

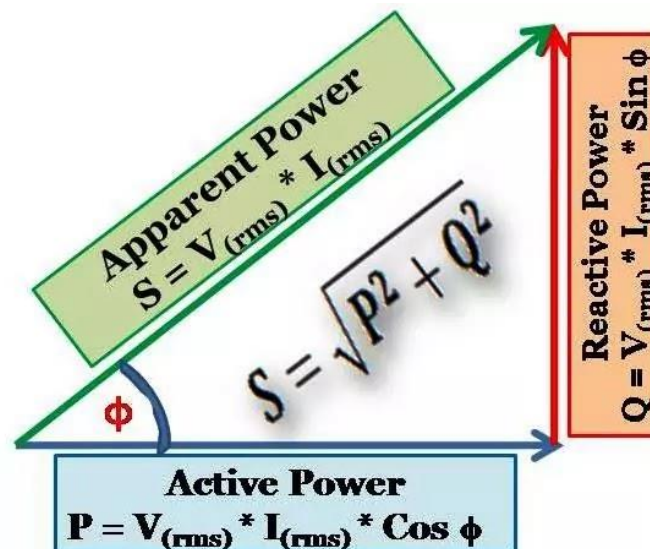
Electrical Loads

- **Domestic Loads:-** Lightning, Fans, Frig, AC , Heater, Oven, Small Motors.
- **Commercial loads:-** Loads at School, Hospital, Hotels, Market places.
- **Industrial Loads:-** Small Scale, Medium Scale & large Scale Industries.
- **Agricultural Loads:-** Motor Pump sets.

Ac Power Review

Apparent Power

- ✓ The combination of reactive power and true power is called apparent power, and it is the product of a circuit's voltage and current, without reference to phase angle.
- ✓ This power is measured in **kVA or MVA**.
- ✓ It has been seen that the power is consumed **only in resistance**. A pure inductor and a pure capacitor do not consume any power,



Real Power

- Alternative words used for Real Power (Actual Power, **True Power**, Watt-full Power, **Useful Power**, and **Active Power**)
- In a DC Circuit, power supply to the DC load is simply the product of Voltage across the load and Current flowing through it i.e., $P = V I$.
- In **DC Circuits, there is no concept of phase** angle between current and voltage. In other words, there is no **Power factor** in DC Circuits

$$P = |S| \cos \phi, \quad P = |V| |I| \cos \phi$$

Measured in watt

Reactive Power

- ✓ Reactive power (Q) is a term for the **imaginary (non-real) power from inductive loads like motor or capacitive loads** (less common)
- ✓ **Reactive power** is used to provide the voltage levels necessary for active **power** to do useful work.
- ✓ **Reactive power** is **essential to move active power** through the **transmission and distribution system to the customer**
- ✓ **Oscillating power** in to and out of the load because of its reactive element (L or C).

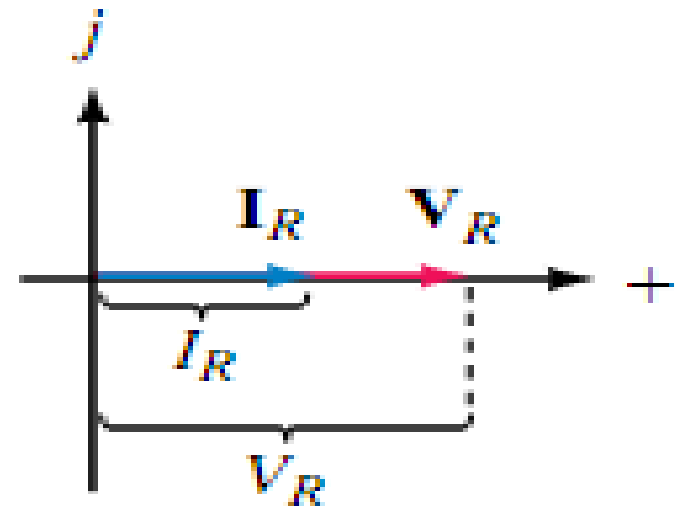
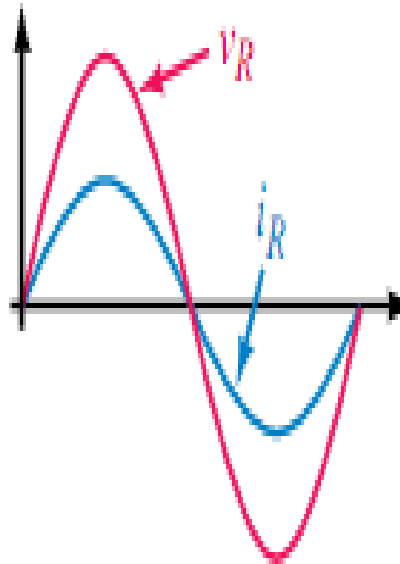
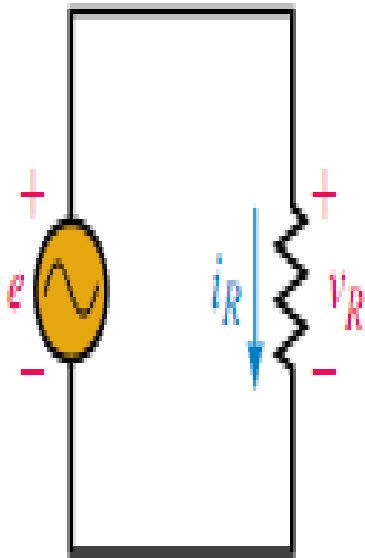
- ✓ Measured in VAR

$$Q = |S| \sin \phi$$

$$Q = |V| |I| \sin \phi$$

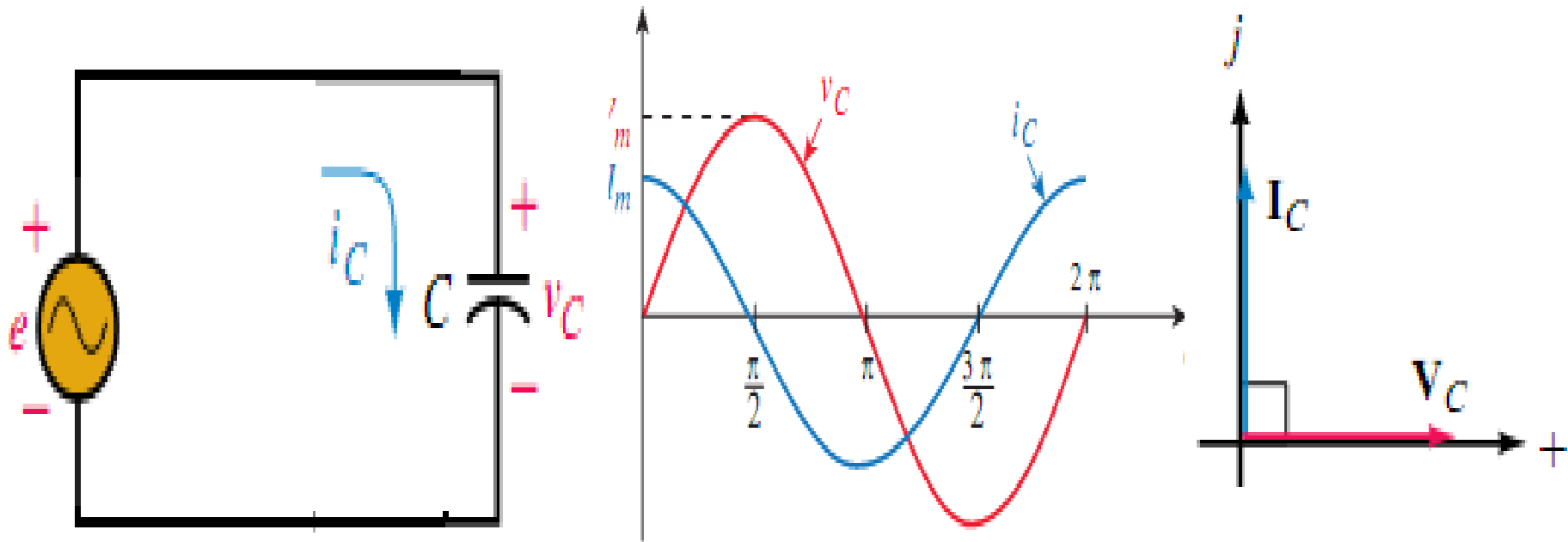
Phase Shifts

- For a pure resistor: the impedance angle is zero,

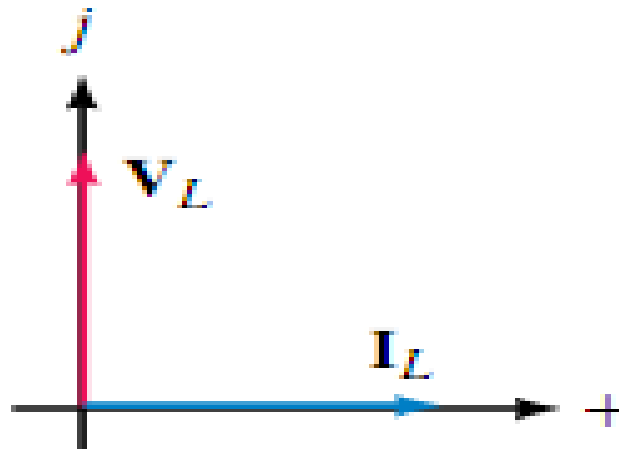
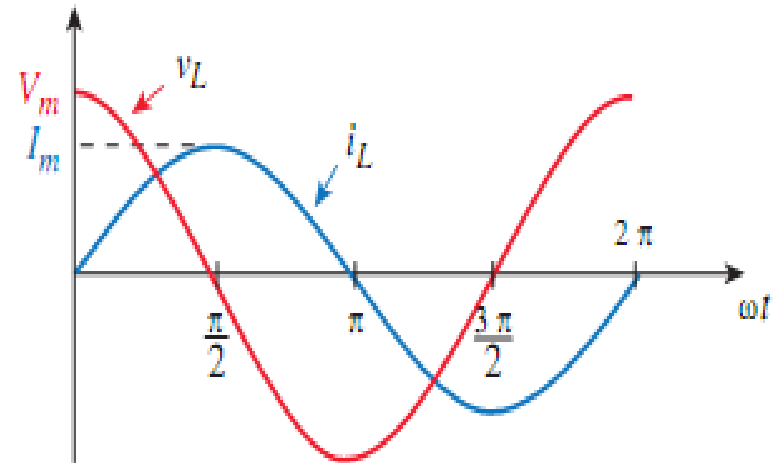
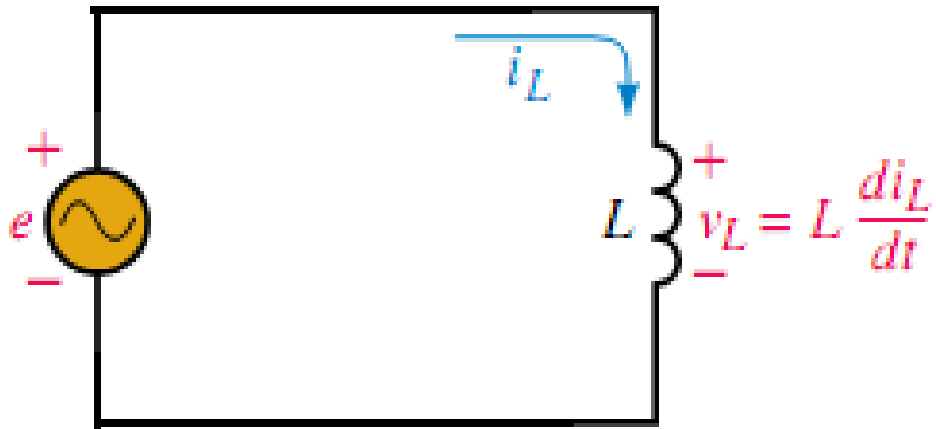


Phase Shifts

- For a purely capacitive : the current leads the voltage by 90°



- purely inductive :- the current lags the voltage by 90° ,



- Real power supplied by the source is equal to the sum of the real powers absorbed by the load and the real losses in the system

$$0 = \sum P_{gen} - \sum P_{loads} - \sum P_{losses}$$

- Reactive power must also be balanced: The balance is between the sum of leading and the sum of lagging reactive power producing elements

$$0 = \sum Q_{leading} + \sum Q_{caps} - \sum Q_{lagging} - \sum Q_{induc}$$

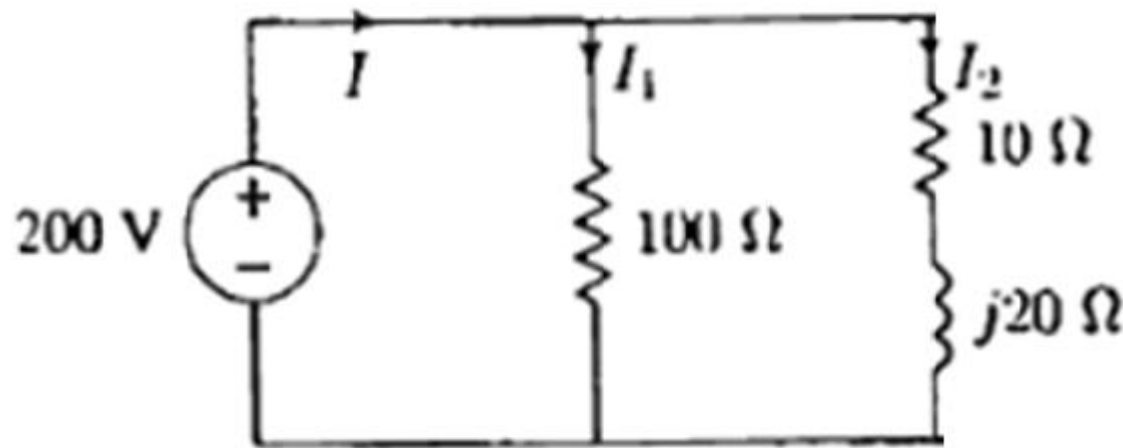
- The total complex power delivered to the loads in parallel is the sum of the complex powers delivered to each

$$0 = \sum S_{gen} - \sum S_{loads} - \sum S_{losses}$$

Example 1

Two loads $Z_1 = 100 \, \Omega$ and $Z_2 = 10 + j20 \, \Omega$ are connected across a 200 V rms 60 Hz source:-

Find the Total real power, reactive Power and power factor of the source.

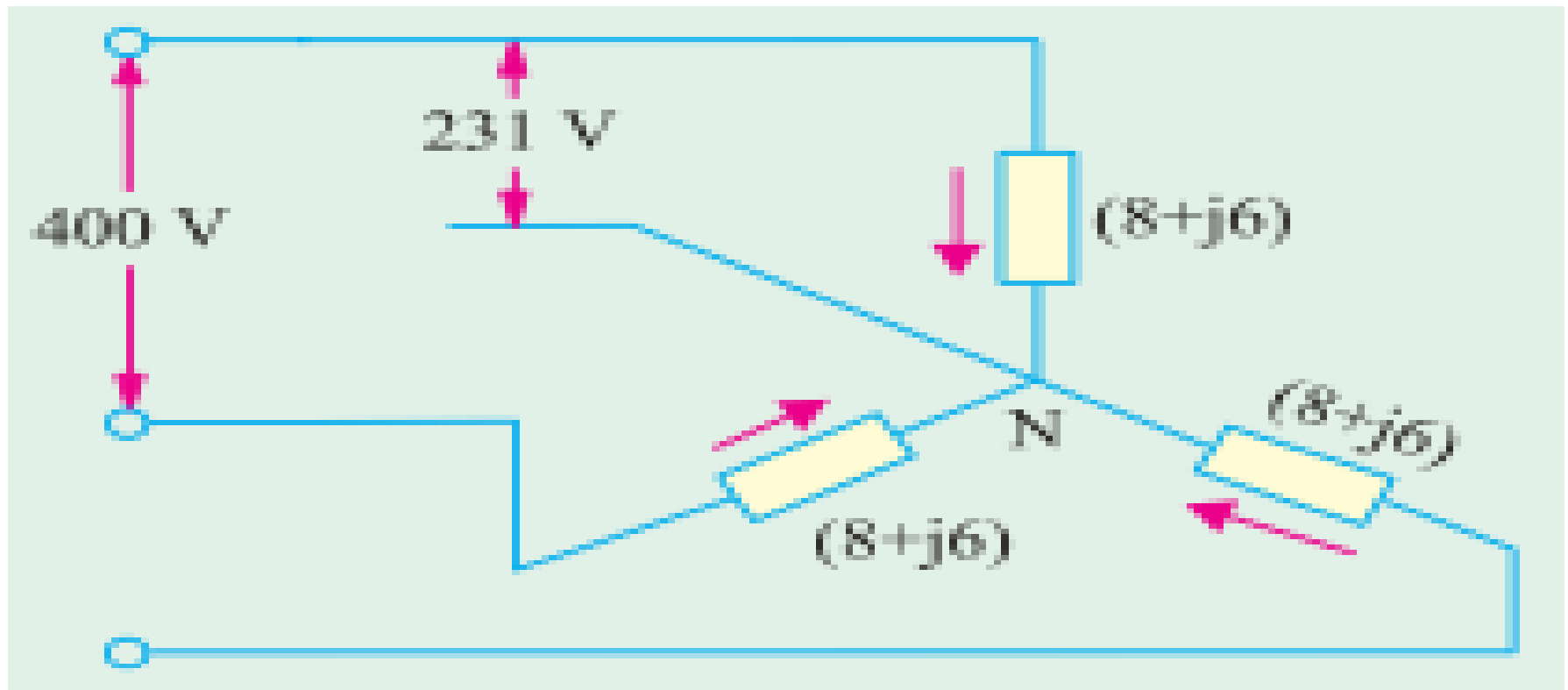


**Answer:- $S = 1200 - j1600 \text{ VA}$
 $\text{pf} = 0.6 \text{ lagging}$**

Example 2:-

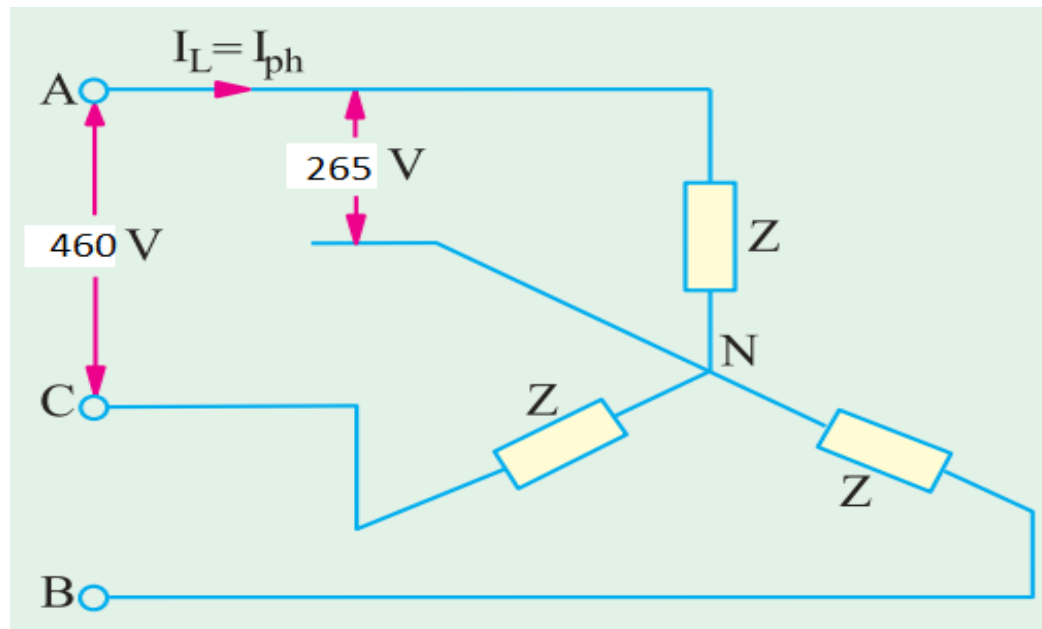
A balanced star connected load of $8+j6\Omega$ per phase, is connected to a balanced 3 phase 400V, supply
Find the line Current, Power factor and total VA.

Answer:- $I_L = 23.1$ A, $Pf = 0.8$ (lagging), $S = 16$ KVA



EXAMPLE 3

Three Equal star connected inductors take 8Kw at power factor of 0.8 when connected across a 460 v,3 – Phase 3-wire system. Find the circuit constants of the load per Phase.



Answer:-

$$R_{ph} = 16.9 \, \Omega$$
$$X_{ph} = 12.66 \, \Omega$$

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End of the Lesson